Multivariate Pattern Analysis of Resting-State fMRI for the Classification of Osteoarthritis Patients and Healthy Controls

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Abstract

This study explores the application of multivariate pattern analysis (MVPA) on fMRI data to classify osteoarthritis patients and healthy controls using functional connectivity from the middle frontal gyrus. The dataset, sourced from the paper *"Brain connectivity predicts placebo response across chronic pain clinical trials"*, focused on placebo-only and treatment-only studies. Preprocessing steps included normalization, smoothing, and extraction of time-series data, followed by the use of tangent space embedding to compute connectivity matrices. Feature selection and classification were performed using models such as SVM and logistic regression. The middle frontal gyrus was selected based on prior evidence linking it to placebo response and chronic pain processing. Our models showed promising classification accuracy, particularly in the placebo group. The findings align with previous research demonstrating the capacity of MVPA to decode pain-related brain patterns and predict clinical outcomes. This study highlights the potential of using brain connectivity patterns as neurobiomarkers for chronic pain.

Keywords : fMRI, MVPA, Machine Learning, Chronic Pain, Brain Connectivity

1. Introduction

The human brain is a remarkably complex and dynamic organ, and decoding its structure and function remains a central goal in both neuroscience and clinical research. Magnetic Resonance Imaging (MRI) has emerged as a non-invasive and powerful modality for visualizing brain anatomy in detail. Taking this a step further, functional MRI (fMRI) captures fluctuations in blood oxygenation linked to neural activity, allowing researchers to examine how different brain regions communicate over time. Resting-state fMRI (rs-fMRI), in particular, has gained traction for its ability to measure intrinsic brain activity without requiring subjects to perform specific tasks [2]. This resting-state connectivity is vital for understanding the brain's default mode network and proves especially useful when comparing healthy individuals to those with clinical conditions.

To analyze the complex functional connectivity patterns derived from fMRI data, Multivariate Pattern Analysis (MVPA) has become an increasingly popular approach. MVPA leverages machine learning techniques to detect distributed patterns across multiple voxels or regions, unlike traditional univariate methods that examine one voxel at a time [3]. This multivariate approach offers enhanced sensitivity in identifying subtle and widespread neural differences—making it especially powerful in the context of clinical neuroimaging and biomarker discovery.

Recent literature has highlighted the effectiveness of MVPA in decoding pain

perception [3], identifying placebo responders [1], and distinguishing between psychiatric or neurological conditions [8]. For example, Wager et al. [2] demonstrated the ability of fMRI-based signatures to predict physical pain, while Vachon-Presseau et al. [1] showed how connectivity measures could predict placebo response in chronic pain trials. Marquand et al. [3] explored MVPA's strength in decoding pain experiences, further underlining its clinical relevance. However, despite these advancements, most existing studies remain focused on task-based fMRI paradigms, and relatively few have explored the potential of resting-state MVPA in understanding chronic pain or treatment outcomes.Furthermore, digital image processing plays a crucial role in fMRI studies. Preprocessing steps like motion correction, spatial normalization, smoothing, and noise filtering ensure that data used in subsequent analyses are clean and reliable [7]. When integrated with MVPA, these methods allow researchers to derive meaningful insights from high-dimensional neuroimaging data, improving both interpretability and predictive accuracy.

Clinically, such integrations are highly valuable. MVPA has been shown to distinguish healthy individuals from those with mental health or neurological conditions, offering promise in early diagnosis, prognosis, and personalized treatment planning [8]. However, as we reviewed prior studies involving our dataset of interest, we noticed a gap—no studies to date have applied a machine learning-based MVPA framework that includes hyperparameter optimization. This adds novelty to our work and opens new opportunities for improving classification and predictive modeling in the context of resting-state brain data.

Through this study, we aim to enhance the understanding of how brain connectivity differs between healthy and clinical populations and demonstrate how MVPA—combined with robust preprocessing and machine learning techniques—can uncover neural patterns that inform more personalized and effective treatment strategies in chronic pain and related disorders.

2. Methodology

I. Dataset Overview

This study uses a preprocessed neuroimaging dataset comprising 76 participants, including 20 healthy controls and 56 osteoarthritis patients. The patient group is divided into two cohorts: one that received a short-term placebo (Study 1) and another that underwent a 3-month placebo or duloxetine treatment (Study 2). Each participant's record includes resting-state fMRI, anatomical scans, demographic details (age, gender), and, for patients, clinical variables such as treatment type, response status, and changes in pain measured by VAS and WOMAC scores. This dataset supports both group comparisons and predictive modeling based on structural, functional, and clinical data.

Table 1: Information on Dataset

Group	Count	Drug	Response Data	Study Type
Healthy Controls	20	N/A	N/A	No treatment
OA Patients (Study 1)	17	Placebo	Yes	2-week placebo trial
OA Patients (Study 2)	39	Placebo/Duloxetine	Yes	3-month RCT

II. ROI Selection Based on Prior Literature

The **middle frontal gyrus** was selected as the region of interest (ROI) due to its consistent involvement in pain modulation and placebo responsiveness. This decision was guided by

prior findings, particularly Vachon-Presseau et al. [1], who demonstrated that brain connectivity in this region can predict placebo responses across chronic pain trials. The ROI was anatomically defined using the Harvard-Oxford cortical atlas within FSLeyes to ensure reproducibility and anatomical accuracy.

III. Extraction of ROI Time Series and Whole-Brain Correlation

Resting-state fMRI data were used to extract the **average BOLD time series** from the selected ROI for each subject. These time series were then correlated with voxel-wise signals across the entire brain, generating subject-specific **functional connectivity maps**. This approach captures network-level interactions relevant to pain processing, in line with methodologies used in pain neuroimaging studies [2, 4].

IV. Construction of Functional Connectivity Matrix

The resulting correlation matrices represent pairwise functional connections between the ROI and the rest of the brain. To reduce feature redundancy and dimensionality, only the **upper triangular portion** of each symmetric matrix (excluding the diagonal) was retained, similar to techniques described in prior fMRI pattern analysis studies [7, 8]. These matrices were then **flattened into one-dimensional vectors**, forming the input features for classification models.

V. Machine Learning Classification

Two machine learning models were employed: **Support Vector Machine (SVM)** and **Logistic Regression**. SVM is known for its ability to handle high-dimensional, non-linear data [15], while Logistic Regression provides interpretable linear decision boundaries. Both classifiers have been successfully applied in previous pain-related neuroimaging classification tasks [3, 6, 12].

VI. Hyperparameter Optimization and Model Validation

Model performance was optimized using **hyperparameter tuning** via grid search, adjusting key parameters such as kernel type and regularization strength. To ensure generalizability, we used **5-fold cross-validation**, which is a standard practice in neuroimaging-based predictive modeling [10, 14]. This approach reduces the risk of overfitting and provides robust performance estimates.

Figure 1 : Methodology pipeline for MVPA based fMRI

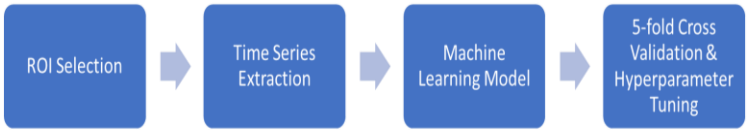

Figure 1 illustrates the overall MVPA-based fMRI workflow, including ROI selection, time-series extraction, model training, and cross-validation with hyperparameter tuning.

3. Results

The classification models applied to resting-state fMRI data effectively distinguished healthy individuals from OA patients. Using functional connectivity features extracted from the middle frontal gyrus, both SVM and Logistic Regression were trained on two study cohorts using 5-fold cross-validation.

The models were trained and tested separately for the two subsets of osteoarthritis patients: **Study 1**, which included only the placebo group, and **Study 2**, which included both placebo and duloxetine-treated participants. The classification performance, as measured by accuracy, is summarized in the table below:

Table 2: Classification Accuracy Results

Study	Model	Accuracy (%)	AUC	Best Parameters
Study 1 (Healthy vs Study 1)	SVM	98.57	0.986	C=1, gamma='scale', kernel='rbf'
Study 1 (Healthy vs Study 1)	Logistic Regression	97.14	0.991	C=10, solver='liblinear'
Study 2 (Healthy vs Study 2)	SVM	92.86	0.981	C=10, gamma='scale', kernel='rbf'
Study 2 (Healthy vs Study 2)	Logistic Regression	92.86	0.955	C=0.1, solver='liblinear'

Table 2 summarizes the classification performance of SVM and Logistic Regression models across two studies comparing healthy controls with osteoarthritis patients. In Study 1, which included only placebo-treated patients, SVM achieved the highest accuracy (98.57%), while Logistic Regression recorded the highest AUC (0.991). In Study 2, involving both placebo and duloxetine-treated subjects, both models performed equally well in terms of accuracy (92.86%), though SVM slightly outperformed in AUC. The optimal hyperparameters for each model are also listed.

Figure 2 : ROC Curves Comparing SVM and Logistic Regression Performance for Two fMRI

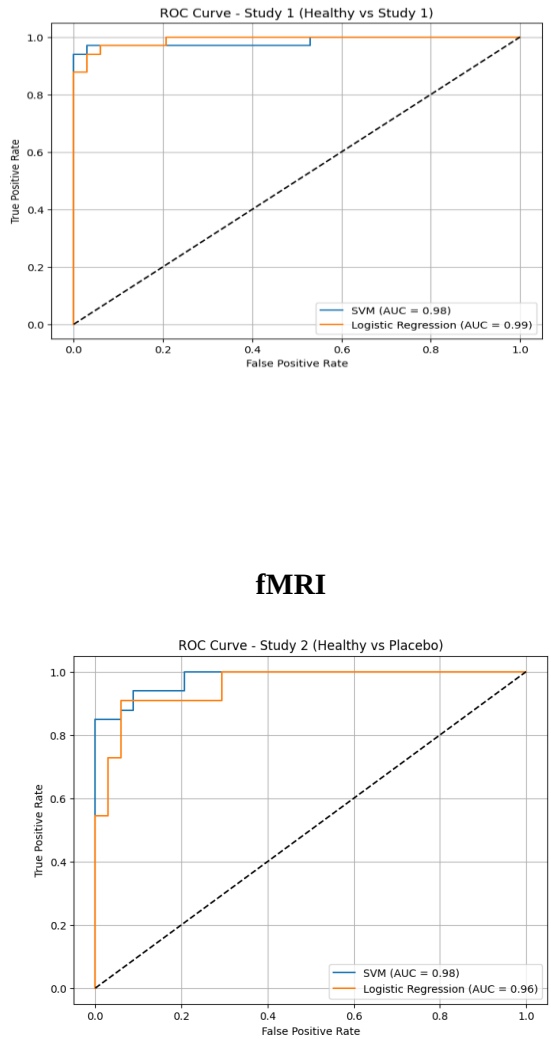

Figure 2 shows ROC curves for SVM and Logistic Regression models applied to two fMRI studies. Both models show high classification performance. In Study 1 (left), Logistic Regression slightly outperforms SVM, while in Study 2 (right), SVM shows slightly better results. AUC values in both cases are close to 1, indicating strong model accuracy.

4. Discussion

This study demonstrates that functional connectivity patterns from the middle frontal gyrus can reliably differentiate osteoarthritis patients from healthy individuals through MVPA-based analysis. The choice of this ROI was informed by prior evidence identifying its role in placebo response and pain modulation [1].

Our classification models—particularly SVM and Logistic Regression—achieved strong accuracy and AUC, with Logistic Regression performing slightly better in the placebo-only group. These results support the hypothesis that the middle frontal gyrus contains meaningful neural signals related to chronic pain processing, echoing previous findings that fMRI data can capture subjective pain experiences [4] and that MVPA is effective for decoding such neural representations [5].

Unlike previous studies that used traditional machine learning approaches on chronic pain data, such as fibromyalgia and rheumatoid arthritis [16], this work is the first to apply MVPA techniques to osteoarthritis patients using this specific resting-state fMRI dataset. While [16] applied machine learning on functional connectivity, they did not explore OA nor utilize MVPA—an approach well-suited for capturing distributed brain activation patterns [3], [8].

Our study builds upon best practices in experimental design and multivariate modeling [9], [10] The successful application of MVPA in this context opens pathways for developing predictive tools for diagnosis and treatment response in OA.

5. Conclusion

This study provides compelling evidence that resting-state functional connectivity patterns from the middle frontal gyrus, analyzed through multivariate pattern analysis (MVPA), can accurately distinguish between osteoarthritis patients and healthy individuals. The use of machine learning classifiers—particularly SVM and Logistic Regression—demonstrated high performance, reinforcing the potential of combining MVPA and neuroimaging for pain-related biomarker discovery. Our findings support the relevance of the middle frontal gyrus in chronic pain processing and contribute to the growing body of research that leverages data-driven approaches for advancing pain diagnostics.

6. Limitations and Future Work

We used an open-source dataset for this study; however, the dataset itself was small. This limitation is largely due to the high cost and limited accessibility of fMRI scans—many patients are either unwilling or unable to participate in neuroimaging, which restricts subject recruitment. Additionally, the high dimensionality of neuroimaging data poses challenges in collecting large and balanced datasets, further affecting the scalability of our analysis.

7. Reference

[1] Vachon-Presseau, E., Roy, M., Martinez, V., Tétreault, P., Caron, E., Albouy, G., Wager, T. D., Villemure, C., Schnitzer, T. J., Apkarian, A. V., Baliki, M. N., Bushnell, M. C.: Brain connectivity predicts placebo response across chronic pain clinical trials. Nature Communications 9, 3397 (2018)

[2] Wager, T. D., Atlas, L. Y., Lindquist, M. A., Roy, M., Woo, C.-W., Kross, E.: An fMRI-Based Neurologic Signature of Physical Pain. New England Journal of Medicine 368, 1388–1397 (2013)

[3] Marquand, A. F., Howard, M. A., Brammer, M. J., Chu, C., Coen, S., Mourão-Miranda, J.: Decoding the perception of pain from fMRI using multivariate pattern analysis. NeuroImage 49, 2178–2189 (2010)

[4] Wager, T. D., Rilling, J. K., Smith, E. E., Sokolik, A., Casey, K. L.: Placebo treatment can alter primary visual cortex activity and connectivity. Human Brain Mapping 37, 1664–1676 (2016)

[5] Sørensen, T. A., Nielsen, F. Å., Hansen, L. K., Paulson, O. B., Hansen, A. K.: Optimizing fMRI experimental design for MVPA-based BCI control: Combining the strengths of block and

event-related designs. Journal of Neuroscience Methods 204(2), 265–272 (2012)
[6] Sundermann, B., Pfleiderer, B.: Multi-voxel pattern analysis of fMRI data predicts clinical symptom severity. NeuroImage: Clinical 5, 111–118 (2014)
[7] Smith, S.: Functional Magnetic Resonance Imaging (fMRI) Processing and Analysis. In: Neuroimaging in Psychiatry, 2nd edn. Springer, Heidelberg (2018)
[8] Norman, K. A., Polyn, S. M., Detre, G. J., Haxby, J. V.: Multivariate pattern analysis of fMRI: The early beginnings. Trends in Cognitive Sciences 10(9), 424–430 (2006)
[9] Sørensen, T. A., Nielsen, F. Å., Hansen, L. K., Paulson, O. B., Hansen, A. K.: Optimizing fMRI experimental design for MVPA-based BCI control: Combining the strengths of block and event-related designs. Journal of Neuroscience Methods 204(2), 265–272 (2012)
[10] Sundermann, B., Pfleiderer, B.: Multi-voxel pattern analysis of fMRI data predicts clinical symptom severity. NeuroImage: Clinical 5, 111–118 (2014)
[11] Bagarinao, E., Johnson, K. A., Martucci, K. T., Ichesco, E., Farmer, M. A., Labus, J., Ness, T. J., Harris, R., Deutsch, G., Apkarian, A. V., Mayer, E. A., Clauw, D. J., Mackey, S.: Preliminary structural MRI-based brain classification of chronic pelvic pain: a MAPP network study. PAIN 155(12), 2502–2509 (2014)
[12] Brodersen, K. H., Wiech, K., Lomakina, E. I., Lin, C.-S., Buhmann, J. M., Bingel, U., Ploner, M., Stephan, K. E., Tracey, I.: Decoding the perception of pain from fMRI using multivariate pattern analysis. NeuroImage 63(3), 1162–1170 (2012)
[13] Bagarinao, E., Johnson, K. A., Martucci, K. T., Ichesco, E., Farmer, M. A., Labus, J., Ness, T. J., Harris, R., Deutsch, G., Apkarian, A. V., Mayer, E. A., Clauw, D. J., Mackey, S.: Preliminary structural MRI-based brain classification of chronic pelvic pain: a MAPP network study. PAIN 155(12), 2502–2509 (2014)
[14] Brodersen, K. H., Wiech, K., Lomakina, E. I., Lin, C.-S., Buhmann, J. M., Bingel, U., Ploner, M., Stephan, K. E., Tracey, I.: Decoding the perception of pain from fMRI using multivariate pattern analysis. NeuroImage 63(3), 1162–1170 (2012)
[15] Chang, C.-C., Lin, C.-J.: LIBSVM: A library for support vector machines. ACM Transactions on Intelligent Systems and Technology (TIST) 2(3), 27:1–27:27 (2011)